

ML Ops Assignment 2

Spark Preprocessing Pipeline

This was the preprocessing pipeline used (of Spark ML):

```
pipeline_stages = [  
    # handle missing numerical data  
    Imputer(inputCols=numericalCols,  
    outputCols=imputedNumericalCols),  
    # encode categorical data and handle missing categorical data  
    StringIndexer(inputCols=categoricalCols,  
    outputCols=indexedCategoricalCols, handleInvalid="keep"),  
    # Assemble and scale the numerical features (required for model)  
    VectorAssembler(inputCols=imputedNumericalCols,  
    outputCol='num_features_assembled'),  
    StandardScaler(inputCol='num_features_assembled',  
    outputCol='num_features_scaled', withStd=True, withMean=False),  
    # Combine all final (numerical, categorical) into 'features'  
    VectorAssembler(inputCols=['num_features_scaled'] +  
    indexedCategoricalCols, outputCol='features')  
]  
  
preprocessing_pipeline = Pipeline(stages=pipeline_stages)
```

Here each pipeline step creates new columns that get appended to dataframe (when pipeline is run):

- simple imputer handles missing values in numerical columns by replacing them with column mean.
- **StringIndexer** encodes categorical columns by encoding category strings into numerical form. It also handles missing values in categorical columns due to **handleInvalid="keep"**.
- Logistic Regression model requires standardized features, so **VectorAssembler** assembles numerical features into a single column (of type vector), which is then scaled by **StandardScaler**.
- Finally all columns are combined into a *features* column using another **VectorAssembler** so that training script's model can use it to train.

Hyper Parameter Search Space

The following hyper parameters search space was defined (for Logistic Regression model). Here **C = 1/regParam** is inverse of regularization strength.

Total number of hyper parameter combinations searched: $4 * 3 = 12$

```
search_space = {
    "C": tune.choice([1,0,0.1]),
    "max_iter": tune.choice([10,20,50,100]),
}
```

Best Hyper Parameters

Best Hyper Parameters found (copied from Mlflow UI Params) are: $C = 1.0$, $max_iter = 100$.

Final Model Performance

This is copied from Mlflow UI Metrics:

Metric	Value
accuracy	0.8797828438639369
precision	0.7874518915314711
recall	0.8797828438639369
f1_score	0.8235750733889323
roc_auc	0.9171627985245795
training_time	14.00402569770813

Predictions CSV

```
import mlflow
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("GetRegisteredModel").getOrCreate()
model = mlflow.spark.load_model('models:/Assignment2Classifier_DA25M622/1')
test_df = spark.read.parquet("test_processed_DA25M622.parquet")
predictions = model.transform(test_df)
(
    predictions

    .select('id', 'age', 'job', 'marital', 'education', 'default', 'balance', 'housing',
            'loan', 'y')
    .coalesce(1)          # this forces only a single part CSV to be saved
    .write.csv('new_predictions.csv', header=True)
)
```

This saved a folder *predictions.csv* having a single part CSV file *part-00000-d3fad608-2803-4634-aa1e-aab856253f71-c000.csv* . The CSV is submitted alongside this report.

I loaded mlflow registered model (i.e. best model found in hyperparameter search) using URI format `models:{MODEL_NAME}/{VERSION}` and Spark is used to do model inference on the

preprocessed test data parquet.

Mlflow UI Screenshots

mlflow ui was run in a dedicated terminal to start UI at <http://localhost:5000> .

Mlflow Run Best Metrics and Params (hyperparameters):

Overview

Model metrics

System metrics

Traces

Artifacts

Description

No description

Metrics (6)

Metric	Value
accuracy	0.8797828438639369
precision	0.7874518915314711
recall	0.8797828438639369
f1_score	0.8235750733889323
roc_auc	0.9171627985245795
training_time	14.00402569770813

Parameters (2)

Parameter	Value
C	1.0
max_iter	100

About this run

Created at 07/18/2026, 06:13:02 PM

Created by sohang

Experiment ID 1

Status Running

Run ID fb2fbd13b5c946b2b2323b53fde2f005

Duration

Source default_worker.py

Logged models

Registered prompts

Datasets

None

Tags

Add tags

Registered models

Assignment2Classifier_DA25M622 v1

Mlflow Artifacts - here on clicking Model/ , Register Model button shows which was used to register model:

Overview

Model metrics

System metrics

Traces

Artifacts

model

Path: /home/sohang/Projects/iti-madras-web-mech-ai/trimester3/DA5402W_ML/OPS/assignments/Assignment2/mlruns/1/fb2fbd13b5c946b2b2323b53fde2f005/artifacts/model

Register model

Mlflow Model Registry shows the registered model:

Registered Models

Search registered models

Name	Latest version	Aliased versions	Created by	Last modified	Tags
Assignment2Classifier_DA25M622	Version 1		sohang	07/18/2026, 06:25:2...	

New model registry UI

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